PROBLEM SOLUTION FIT

Project Name: EduGenie

Project Type: AI-Powered Learning Assistant

Technology: Google Gemini Generative AI

1. INTRODUCTION

EduGenie is designed to solve the problem of students spending significant time reading, understanding and summarizing lengthy educational content.

The project connects the identified student problem with an AI-based solution using Google Gemini Generative AI.

2. IDENTIFIED PROBLEM

Students commonly face the following challenges:

- Large amounts of educational content.
- Difficulty identifying important information.
- Time-consuming manual note preparation.
- Difficulty understanding complex topics.
- Limited time for examination preparation.
- Repeated reading during revision.

3. PROPOSED SOLUTION

EduGenie provides an AI-powered learning assistant that processes educational content and generates concise and understandable learning responses.

The system uses Google Gemini Generative AI to understand the user's input and generate an appropriate response.

4. PROBLEM–SOLUTION FIT

Problem 1:

Students spend a lot of time reading lengthy content.

Solution:

EduGenie can generate concise summaries from the provided educational content.

Problem 2:

Students find it difficult to identify important information.

Solution:

The AI-generated response can highlight the key information from the provided content.

Problem 3:

Students may find complex educational topics difficult to understand.

Solution:

EduGenie can provide simpler explanations based on the user's learning request.

Problem 4:

Manual note preparation takes time.

Solution:

EduGenie can assist students by generating structured and concise learning material.

Problem 5:

Students need quick revision support.

Solution:

The generated summaries and important points can be used as revision material.

5. VALUE PROVIDED TO USERS

EduGenie provides value by:

- Saving time during study and revision.
- Reducing the effort required for manual summarization.
- Making educational content easier to understand.
- Providing AI-assisted learning support.
- Helping students focus on important information.

6. USER NEED VS SOLUTION

User Need	EduGenie Solution
Quick revision	AI-generated summaries
Easy understanding	Simple AI explanations
Important information	Key information extraction
Less manual work	Automated AI assistance
Learning support	Gemini-powered responses
Simple interaction	Student-friendly interface

7. FEASIBILITY

The proposed solution is technically feasible because:

- Google Gemini provides Generative AI capabilities.
- Python can be used for application development.
- The Gemini API enables communication between the application and AI model.
- The solution can be expanded with additional learning features.

8. EXPECTED IMPACT

EduGenie aims to make educational content easier to process and revise.

The project can help students:

- Reduce the time spent on lengthy study material.
- Understand important concepts more easily.
- Prepare concise revision material.
- Use Generative AI as a learning support tool.

9. FUTURE EXTENSION

The Problem-Solution Fit can be further strengthened by adding:

- AI question answering.
- Quiz generation.
- Flashcard creation.
- Personalized learning.
- Topic-wise revision.
- Student progress tracking.

10. CONCLUSION

EduGenie provides a direct connection between the identified student problems and the proposed AI solution. By using Google Gemini Generative AI, the project addresses challenges related to lengthy educational content, manual summarization, understanding and revision.

Therefore, EduGenie provides a practical foundation for an AI-assisted student learning solution.